DATA SCIENCE REPORT

PRODUCT TAXONOMY INTELLIGENCE: NLP-BASED PRODUCT CLASSIFICATION & EXPLAINABILITY

Character-Level Feature Engineering, Machine Learning & Classification Performance



TOOLS USED

Python, Pandas, Numpy, Matplotlib, Plotly & ML libraries

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Overview & Business Objective

1.1 PROJECT OVERVIEW

E-commerce platforms and product comparison services manage large catalogs containing thousands of product listings from different merchants. These listings often vary in naming conventions, technical specifications, brand references, and product descriptions.

Manually categorizing such products can be time-consuming and inconsistent, particularly when similar products are listed under different names.

The Product Taxonomy Intelligence project applies Natural Language Processing (NLP) and machine learning to automatically classify products into predefined categories using their titles.

The project covers data exploration, text preprocessing, TF-IDF feature engineering, classical machine learning, TextCNN experimentation, model evaluation, error analysis, and explainability.



1.2 PROBLEM STATEMENT

Product titles frequently contain abbreviations, technical specifications, model identifiers, and overlapping terminology, making accurate classification challenging.

The key problem addressed in this project is:

How can machine learning classify products into predefined categories using product titles while maintaining strong performance across different product types?

The project formulates this as a 10-class supervised text classification problem, with *Category Label* as the target variable.

1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Analyze product listings and identify category-specific text patterns.
- Prepare product titles for NLP-based classification.
- Develop and compare word-level and character-level TF-IDF representations.
- Train and optimize multiple machine learning classifiers.
- Evaluate a TextCNN architecture as a deep learning alternative.
- Prevent information leakage through group-aware data splitting and cross-validation.
- Assess classification performance and investigate prediction errors.
- Interpret model decisions using character-level feature contributions.



1.4 PROJECT SCOPE

The project follows a structured workflow covering data analysis, NLP-based feature engineering, machine learning classification, model explainability, and business interpretation to identify product naming patterns, evaluate classification performance, understand model decisions, and explore opportunities for automated product categorization.

01

Data Analysis

- Dataset inspection
- Data quality assessment
- Category distribution analysis
- Merchant and cluster analysis

02

NLP & Classification

- Text preprocessing
- TF-IDF feature extraction
- Group-aware validation
- Machine learning & TextCNN modeling

03

Model Explainability

- Character feature importance
- Misclassification analysis
- Decision margin analysis

04

Business Interpretation

- Automated product categorization
- Product catalog consistency
- Ambiguous listing identification
- Classification limitations
- Recommendations for improvement

Data Understanding & Exploratory Data Analysis

2.1 DATA OVERVIEW

The dataset provides a comprehensive view of product listings across multiple merchants and categories, covering product titles, identifiers, category labels, and product clustering information. These variables provide the foundation for analyzing product naming patterns, identifying category-specific terminology, and developing NLP-based machine learning models for accurate product classification.

Data Source: Kaggle

Dataset Size

35,311
Records

16
Original Features

Data Diversity

10
Categories

306
Merchamts

12,849
Clusters

DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Product ID	Categorical (Identifier)	Unique listing identifier
Product Title	Text (Unstructured)	Textual product name
Merchant ID	Categorical (Identifier)	Identifier of the merchant
Cluster ID	Categorical (Identifier)	Product grouping identifier
Cluster Label	Text (Categorical Label)	Product grouping description
Category ID	Categorical (Nominal)	Numerical product category identifier
Category Label	Categorical (Nominal)	Product category name

2.2 TARGET VARIABLE

The target variable selected for prediction is:

Demand

Category Label represents the product category that the model attempts to predict based on the textual information contained in each product title.

The target has:

10
Unique Categories

2,212
Smallest Category Count

5,501
Largest Category Count

15.58%
Largest Category Share

6.26%
Smallest Category Share

The distribution is moderately imbalanced, with Fridge Freezers representing the largest category and Freezers the smallest.

2.3 DATA QUALITY & PREPARATION OVERVIEW

Before modeling, the dataset was examined for:

- Data types
- Missing values
- Duplicate observations
- Categorical variables
- Numerical variables
- Temporal information
- Potential outliers
- Feature relationships

The assessment identified:

- 35,311 observations and seven variables.
- No missing values across the original columns.
- No fully duplicated rows.
- 30,993 unique product titles, indicating repeated titles across listings.

Following data preparation, 35,310 observations were retained for modeling.

Repeated product titles and product clusters were important considerations when designing the train-test split to reduce information leakage.

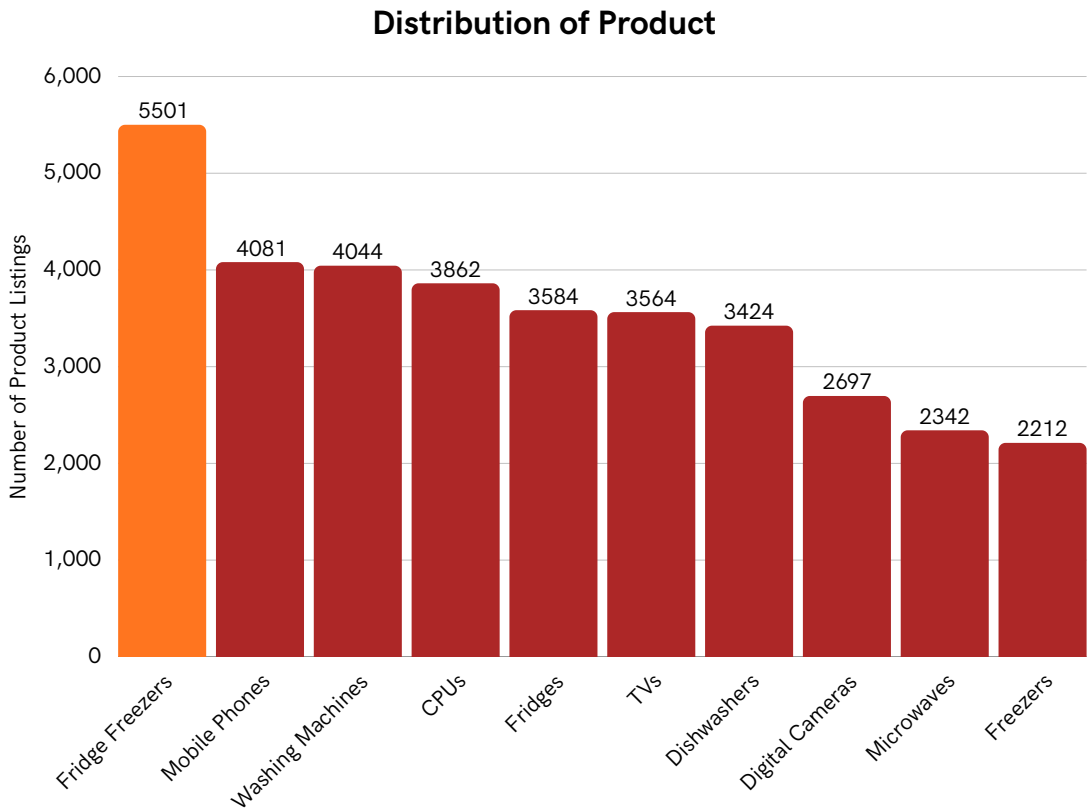


2.4 Exploratory Data Analysis

Product categories are moderately imbalanced, with Fridge Freezers representing the largest share of listings.

15.58%

Fridge Freezer Listings



Key observations

- Fridge Freezers account for approximately 15.58% of listings, representing the largest category.
- Freezers account for approximately 6.26%, making them the least represented category. The remaining categories have relatively moderate differences in representation.

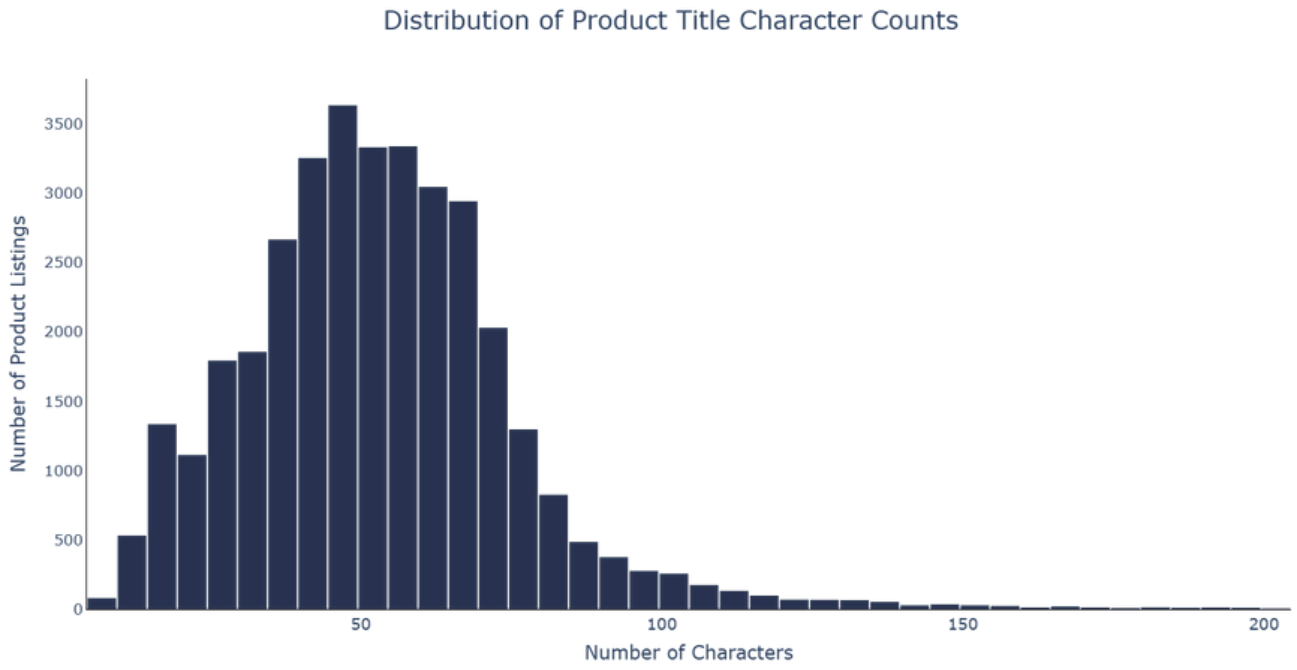
Business Insights

- Category-level evaluation is necessary to identify performance differences that overall accuracy may conceal.
- Macro-averaged metrics provide a useful assessment of classification performance across categories.

Product titles vary considerably in length, averaging approximately 53 characters and nine words per listing.

53

Average Title Length



Key observations

- Product titles contain approximately 53 characters and 8.67 words on average.
- Title lengths range from 5 to 200 characters.
- Product descriptions vary considerably in the amount of information provided.

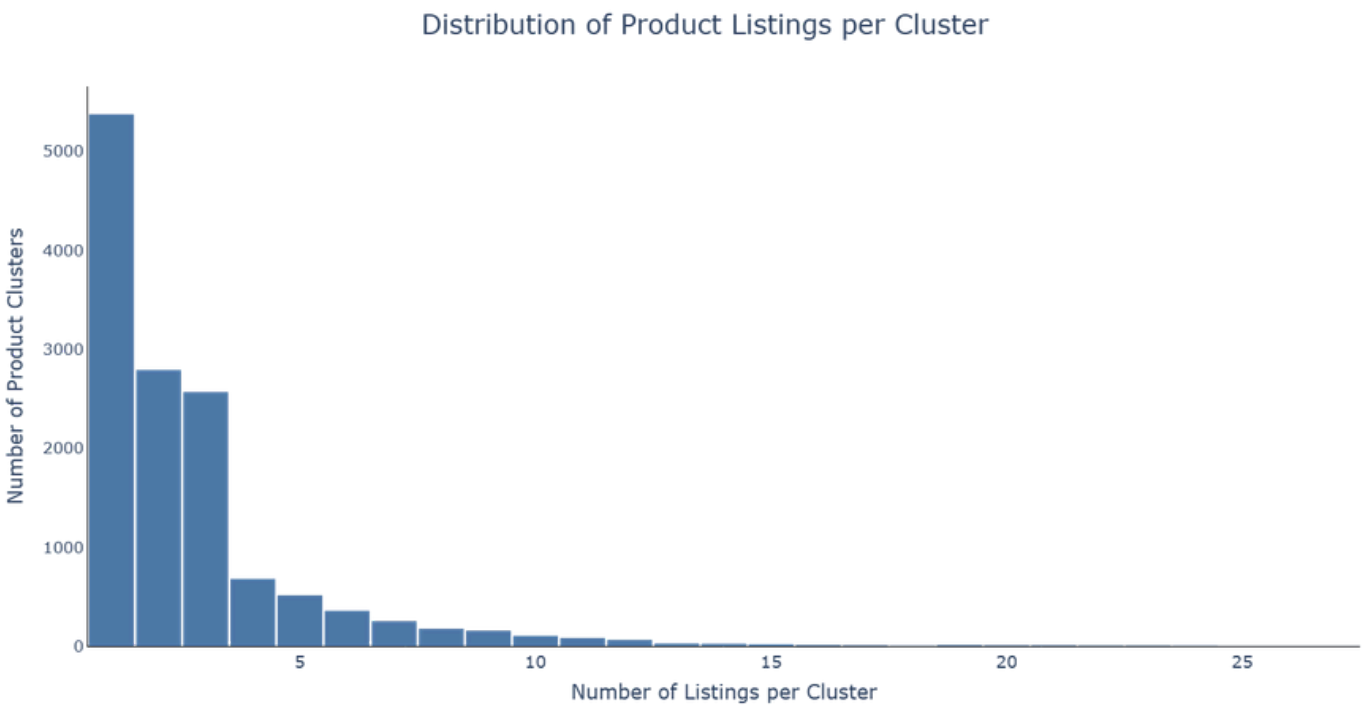
Business Insights

- Short titles may provide limited evidence for classification.
- Character-level representations can capture useful patterns within abbreviated names and model identifiers.

Most product clusters contain multiple listings, indicating repeated representations of underlying products.

59.39%

Multi-Listing Cluster Share



Key observations

- The dataset contains 13,233 unique product clusters. Of these, 7,859 clusters contain multiple listings.Individual clusters contain up to 27 listings.

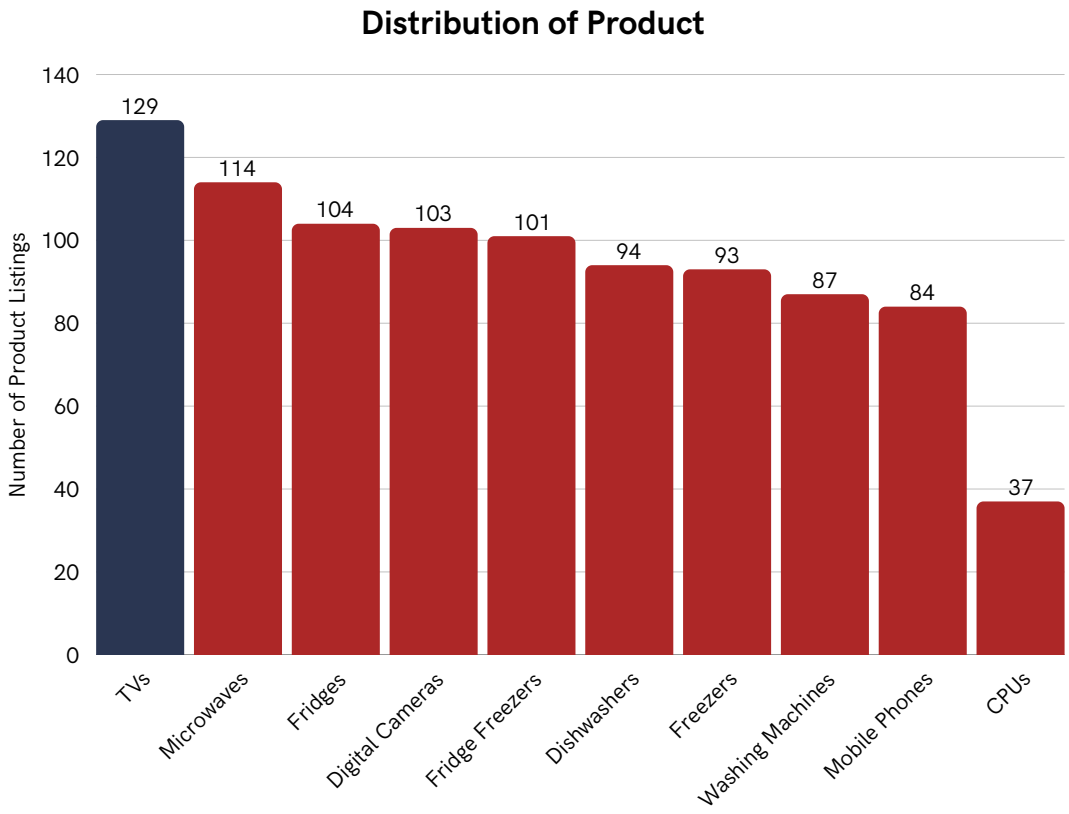
Business Insights

- Multiple listings may refer to the same underlying product.
- Product grouping information must be considered during validation to avoid overly optimistic performance estimates.

Product listings are unevenly distributed across merchants, with the largest merchant contributing substantially more listings.

2,547

Largest Merchant Listings



Key observations

- The dataset contains 306 unique merchants. Merchant representation varies substantially, with the largest merchant contributing 2,547 listings.
- Some merchants specialize in a single category, while others list products across multiple categories.

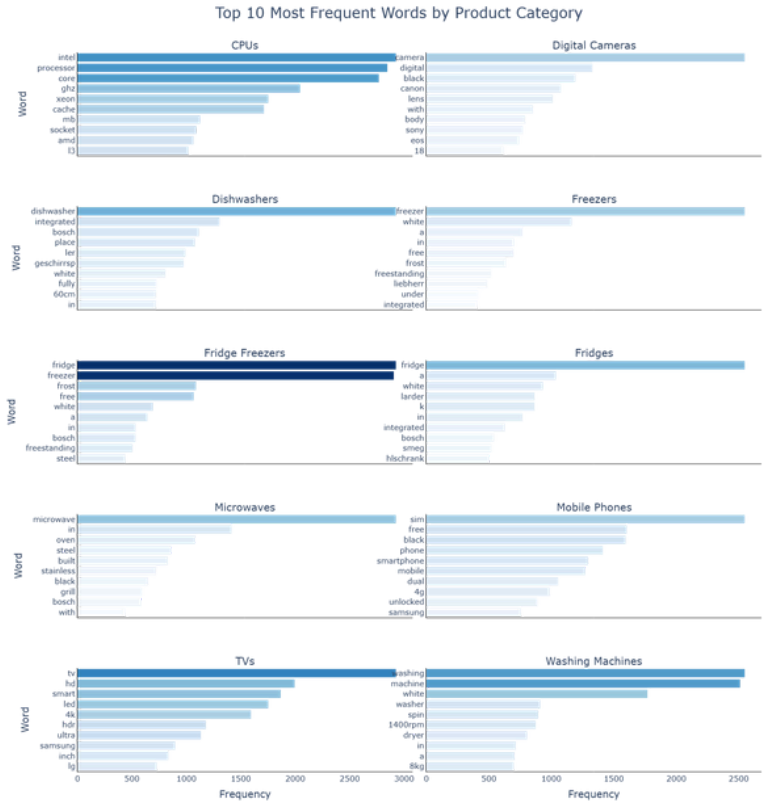
Business Insights

- Product naming patterns may vary between merchants.
- A robust classification system should learn transferable product terminology rather than depend exclusively on merchant-specific patterns.

Product categories exhibit distinctive terminology, although related appliance categories share overlapping keywords.

10

Product Categories Analyzed



Key observations

- Product categories exhibit recognizable terminology associated with their functions and technical specifications.
- Common terms include fridge, freezer, intel, processor, washing, and smart.
- Related categories share terminology, particularly Fridges, Freezers, and Fridge Freezers.

Business Insights

- Text-based classification is feasible because product titles contain meaningful category information.
- Overlapping terminology may introduce ambiguity between closely related categories.

Data Preparation & Feature Engineering



3.1 DATA PREPARATION

Following exploratory analysis, the dataset was prepared for supervised text classification.

The preparation process included:

- Cleaning and standardizing the dataset.
- Selecting Product Title as the primary predictive input.
- Separating the target variable.
- Identifying related product listings.
- Creating group-aware training and testing partitions.
- Encoding target categories.
- Transforming product titles into numerical representations.

The resulting dataset contained 35,310 observations.



3.2 FEATURE SELECTION & LEAKAGE PREVENTION

The model uses product titles as its primary input rather than relying on product identifiers, merchant identifiers, or category-related metadata.

Identifiers and grouping information were considered during data preparation to reduce leakage between related product listings.

This approach encourages the classifier to learn patterns from product names rather than memorize dataset-specific identifiers.



3.3 TRAIN-TEST SPLIT

The dataset was divided using a group-aware strategy.

Dataset	Observations
Training	28,919
Testing	6,391
Total	***35,310***

The test dataset was reserved for independent evaluation.

Group-aware splitting was used to reduce the risk of closely related product listings appearing across training and testing partitions.

3.4 TEXT FEATURE ENGINEERING

Product titles were transformed into numerical representations using ***Term Frequency–Inverse Document Frequency (TF-IDF)***.

The experimentation considered word-level, character-level, and combined feature representations.

Word-Level TF-IDF

Captures individual words and word combinations that describe product categories, brands, and specifications.

Character-Level TF-IDF

Captures shorter character sequences within product titles.

This representation is particularly useful for:

- Model identifiers and technical specifications.
- Abbreviated product names.
- Variations in product naming conventions.
- Partial words and recurring character patterns.

Combined TF-IDF

Combines word-level and character-level information to evaluate whether complementary representations improve classification.



3.5 TARGET ENCODING

The 10 product categories were encoded into numerical labels for model training and evaluation.

The fitted label encoder was retained to ensure consistent category mappings across subsequent notebooks.

3.6 FINAL FEATURE REPRESENTATION

The selected model uses 50,000 character-level TF-IDF features.

Component	Final configuration
Input	Product Title
Target	Category Label
Classification classes	10
Feature representation	Character TF-IDF
Feature dimensions	50,000
Training observations	28,919
Testing observations	6,391



Model Development & Experimentation

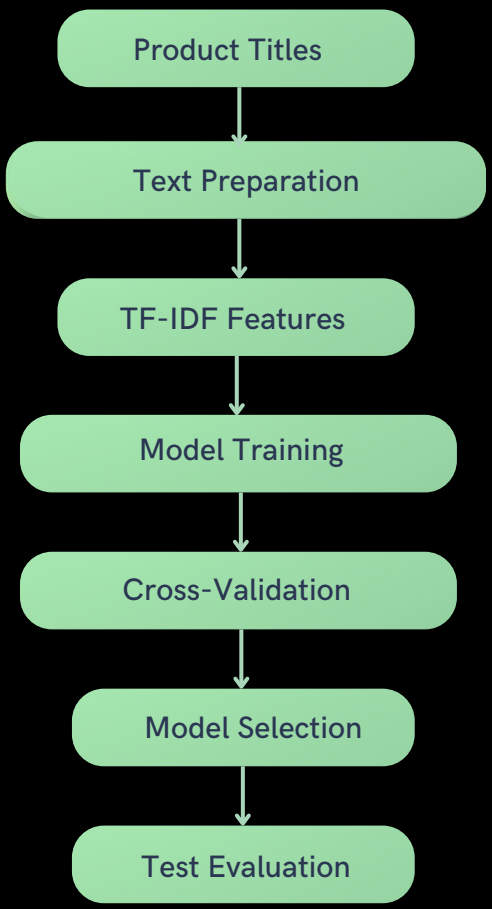


4.1 MODELING APPROACH

The project was formulated as a supervised multiclass text classification problem.

Multiple algorithms and text representations were explored to identify a model offering strong predictive performance and practical computational efficiency.

The development workflow followed:



4.2 CLASSICAL MACHINE LEARNING

The experimentation included classical machine learning approaches suited to sparse text features, including linear classifiers and alternative classification methods.

Model selection considered predictive performance, consistency across validation folds, and computational requirements.

4.3 CROSS-VALIDATION STRATEGY

Model development used 5-fold StratifiedGroupKFold cross-validation.

This approach was selected to:

- Preserve category representation across validation folds.
- Keep related product groups separated.
- Reduce information leakage during model comparison.
- Assess model performance across multiple validation partitions.

Macro F1-score was used as a primary comparison metric to account for differences in category representation.



4.4 MODEL COMPARISON

A comprehensive comparison was conducted across 31 model configurations, including a majority-class baseline, to assess the effectiveness of different classification algorithms and text feature representations.

The experiments evaluated Word TF-IDF, Character TF-IDF, and Combined TF-IDF using three performance metrics:

- Accuracy: Overall proportion of correctly classified product listings.
- Macro F1-score: Classification performance averaged equally across all ten product categories.
- Weighted F1-score: Classification performance adjusted for category representation.

Classical Machine Learning Model Comparison

Model	Feature Representation	Accuracy	Macro F1	Weighted F1
Ridge Classifier	Character TF-IDF	0.9878	0.9877	0.9878
Stacking Classifier	Character TF-IDF	0.9872	0.9871	0.9872
Ridge Classifier	Combined TF-IDF	0.9866	0.9864	0.9866
Voting Classifier	Character TF-IDF	0.9862	0.9860	0.9862



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Model	Feature Representation	Accuracy	Macro F1	Weighted F1
Linear SVM	Character TF-IDF	0.9860	0.9857	0.9860
MLP Classifier	Character TF-IDF	0.9847	0.9845	0.9847
MLP Classifier	Combined TF-IDF	0.9845	0.9843	0.9845
Linear SVM	Combined TF-IDF	0.9840	0.9837	0.9840
SGD Classifier	Character TF-IDF	0.9822	0.9822	0.9822
SGD Classifier	Combined TF-IDF	0.9823	0.9822	0.9823
Logistic Regression	Character TF-IDF	0.9762	0.9763	0.9762
Logistic Regression	Combined TF-IDF	0.9753	0.9752	0.9753
Complement Naive Bayes	Combined TF-IDF	0.9624	0.9582	0.9614
Ridge Classifier	Word TF-IDF	0.9530	0.9558	0.9532
Linear SVM	Word TF-IDF	0.9525	0.9550	0.9526
SGD Classifier	Word TF-IDF	0.9506	0.9524	0.9509
MLP Classifier	Word TF-IDF	0.9491	0.9510	0.9493
Logistic Regression	Word TF-IDF	0.9465	0.9488	0.9467
Multinomial Naive Bayes	Combined TF-IDF	0.9487	0.9433	0.9470
Complement Naive Bayes	Character TF-IDF	0.9517	0.9429	0.9493
Random Forest	Character TF-IDF	0.9353	0.9384	0.9372

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Model	Feature Representation	Accuracy	Macro F1	Weighted F1
Random Forest	Combined TF-IDF	0.9323	0.9364	0.9347
Extra Trees	Character TF-IDF	0.9326	0.9358	0.9347
Complement Naive Bayes	Word TF-IDF	0.9380	0.9350	0.9375
Multinomial Naive Bayes	Character TF-IDF	0.9422	0.9332	0.9393
Extra Trees	Combined TF-IDF	0.9280	0.9303	0.9294
Multinomial Naive Bayes	Word TF-IDF	0.9219	0.9198	0.9215
Random Forest	Word TF-IDF	0.8757	0.8706	0.8752
Extra Trees	Word TF-IDF	0.8066	0.7975	0.8100
Majority Class Baseline	Word TF-IDF	0.1580	0.0273	0.0431

Key Observations

- Ridge Classifier with Character TF-IDF achieved the highest initial Macro F1-score of 0.9877, outperforming the other evaluated configurations.
- Stacking and Voting classifiers delivered competitive performance but did not surpass the strongest individual model.
- Character-level TF-IDF generally outperformed word-level representations, particularly among linear classifiers.
- Tree-based models, including Random Forest and Extra Trees, showed comparatively weaker performance, especially with Word TF-IDF.
- The majority-class baseline achieved a Macro F1-score of only 0.0273, demonstrating the substantial improvement achieved through supervised classification.

Final Model Comparison After Tuning

Following the initial experiments, selected high-performing models were compared using cross-validation results, including tuned configurations.

Model	Feature Representation	CV Macro F1
Ridge Classifier (Tuned)	Character TF-IDF	0.9879
Ridge Classifier	Character TF-IDF	0.9877
Stacking Classifier	Character TF-IDF	0.9871
Voting Classifier	Character TF-IDF	0.9860
Linear SVM	Character TF-IDF	0.9857
Linear SVM (Tuned)	Character TF-IDF	0.9857

Hyperparameter tuning improved Ridge Classifier's Macro F1-score from 0.9877 to 0.9879, while the tuned Linear SVM did not improve upon its initial result.

The results confirmed Ridge Classifier as the strongest-performing classical machine learning approach.

Selected Model Configuration

The experiments evaluated different classification algorithms and feature representations.

The strongest configuration was:

Component	Selected Result
Model	Ridge Classifier
Feature Representation	Character TF-IDF
Regularization Parameter	α = 0.5
Cross-Validation Macro F1	0.9879
Cross-Validation Standard Deviation	0.0021

The selected configuration demonstrated strong and consistent classification performance across validation folds.

Character-level TF-IDF effectively captured product naming patterns, technical specifications, abbreviations, and model identifiers, while Ridge Classifier provided competitive performance without the additional complexity of ensemble methods.

The tuned Ridge Classifier with Character TF-IDF was therefore selected as the final model for evaluation and explainability analysis.

4.5 FINAL MODEL SELECTION

The Ridge Classifier with Character TF-IDF was selected as the final model.

Its advantages included:

- High macro-averaged classification performance.
- Consistent cross-validation results.
- Efficient handling of high-dimensional sparse features.
- Interpretable linear coefficients for feature-level analysis.

Deep Learning Experimentation - TextCNN



5.1 EXPERIMENT OBJECTIVE

A Text Convolutional Neural Network (TextCNN) was evaluated as a deep learning alternative to classical machine learning.

The objective was to determine whether a neural architecture could learn useful local text patterns and improve product category classification.

5.2 TEXTCNN MODELING

TextCNN uses convolutional operations to identify informative patterns within textual sequences.

Unlike TF-IDF-based linear models, the neural architecture learns representations during training.

The experiment was conducted under the project's CPU-based computational constraints.



5.3 CROSS-VALIDATION RESULTS

The TextCNN experiment was evaluated using five-fold cross-validation.

Metric	Result
Cross-validation Macro F1	0.9333
Validation folds	5

The model demonstrated strong classification capability but did not outperform the selected classical machine learning approach.

5.4 CLASSICAL ML VS TEXTCNN

Model	CV Macro F1
Ridge Classifier + Character TF-IDF	0.9879
TextCNN	0.9333

The Ridge Classifier achieved a higher mean cross-validation Macro F1-score by approximately 0.0546.

This suggests that character-level TF-IDF combined with a linear classifier was particularly effective for the product-title classification task.



Final Model Evaluation



6.1 EVALUATION METHODOLOGY

The selected Ridge Classifier was trained on the complete training partition and evaluated against the previously unseen test dataset.

Evaluation included overall classification metrics, category-level performance, confusion matrix analysis, misclassification patterns, product-title length, and decision margins.

6.2 OVERALL CLASSIFICATION PERFORMANCE

Metric	Test result
Accuracy	98.81%
Macro Precision	98.88%
Macro Recall	98.84%
Macro F1-score	98.86%
Weighted F1-score	98.81%

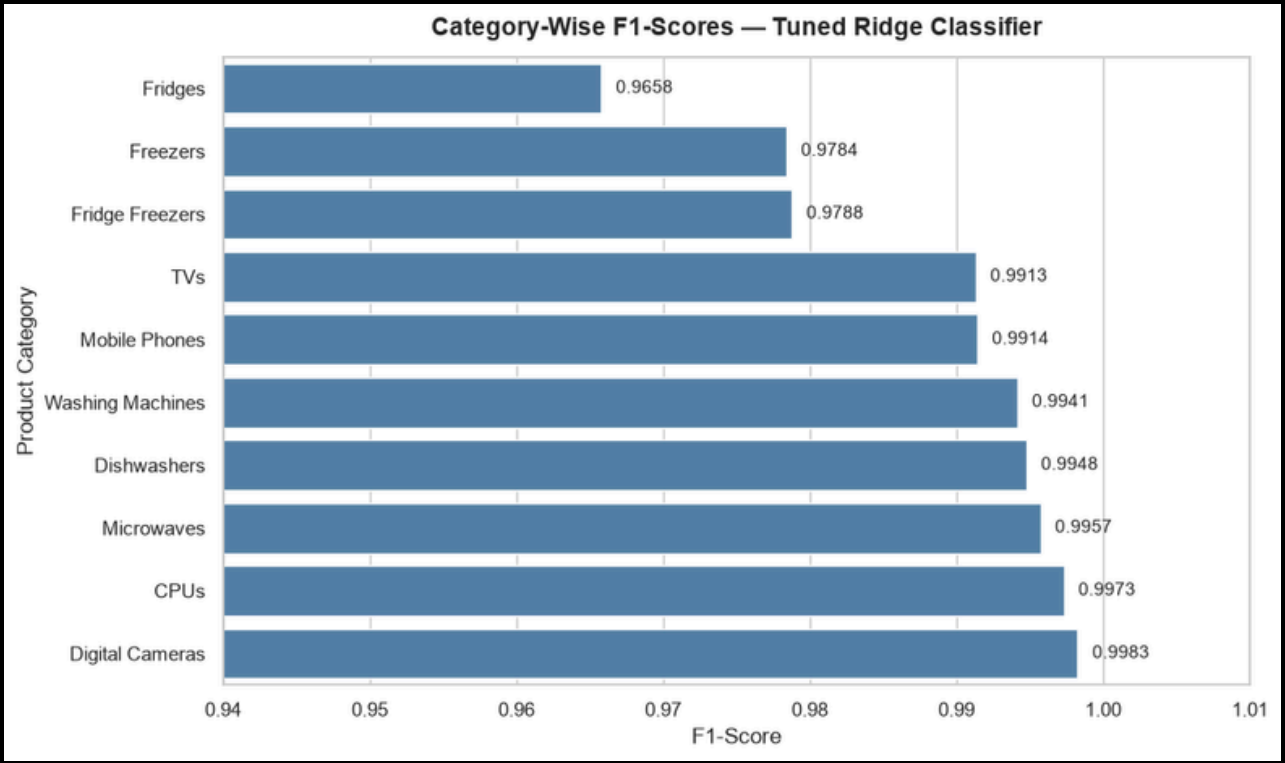
The model correctly classified 6,315 of 6,391 test observations.

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Only 76 predictions were incorrect, corresponding to an error rate of approximately 1.19%.

These results demonstrate strong classification performance across the evaluated product categories.

6.3 CATEGORY-LEVEL PERFORMANCE



The classifier achieved strong performance across all categories, although some closely related product types were more challenging.

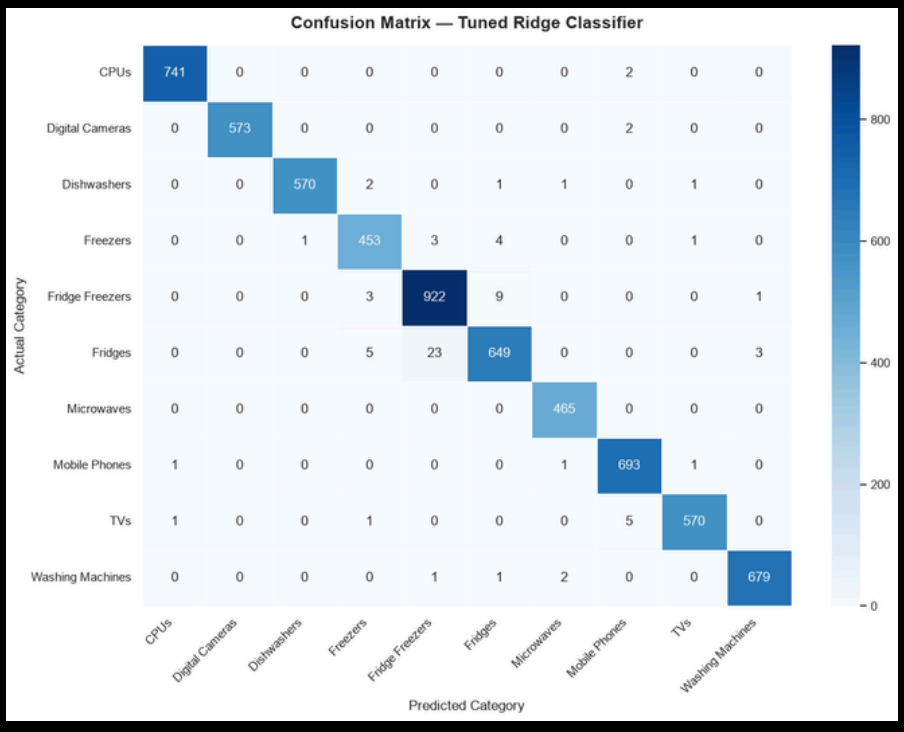
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Category	F1-score
Fridges	0.9658
Freezers	0.9784
Fridge Freezers	0.9788

Fridges recorded the lowest F1-score among the categories.

The results indicate that product categories sharing terminology and functional characteristics can be more difficult to distinguish.

6.4 CONFUSION MATRIX ANALYSIS



The confusion matrix showed that most predictions were concentrated along the main diagonal, indicating a high proportion of correct classifications.

The most notable confusion patterns involved related refrigeration categories:

- Fridges → Fridge Freezers: 23 predictions.
- Fridge Freezers → Fridges: 9 predictions.

These errors suggest that similar product terminology can lead to confusion between closely related categories.

6.5 MISCLASSIFICATION ANALYSIS

The model produced ***76 incorrect predictions***.

Error analysis identified overlapping product terminology and technical specifications as potential sources of ambiguity.

For example, an Android device description containing processor-related term was incorrectly classified as a CPU.

This illustrates how terms such as ***ghz*** and ***core*** can appear across different product categories.



6.6 PREDICTION ERRORS BY PRODUCT TITLE LENGTH

Results obtained:

Product title length	Error rate
1–20 characters	3.16%
21–40 characters	1.98%
41–60 characters	0.62%
61–80 characters	0.90%
81+ characters	1.11%

Key Observations

- Very short product titles recorded the highest error rate.
- Titles containing 41–60 characters achieved the lowest error rate.
- Error rates were not strictly decreasing with title length.

The findings suggest that additional descriptive information can improve classification, although longer titles do not necessarily guarantee better predictions.



6.7 DECISION MARGIN ANALYSIS

Decision margins were examined to understand the separation between the model's highest and second-highest category scores.

Prediction group	Average decision margin
Correct predictions	1.6711
Incorrect predictions	0.3330

Correct predictions generally exhibited substantially larger decision margins.

At an exploratory decision-margin threshold of 1.0:

- Coverage: 90.30%
- Accuracy within retained predictions: 99.91%

These results suggest that decision margins may help identify ambiguous predictions.

However, Ridge Classifier decision scores are not calibrated probabilities, and this threshold was not established as a production decision rule.



Model Explainability



7.1 EXPLAINABILITY APPROACH

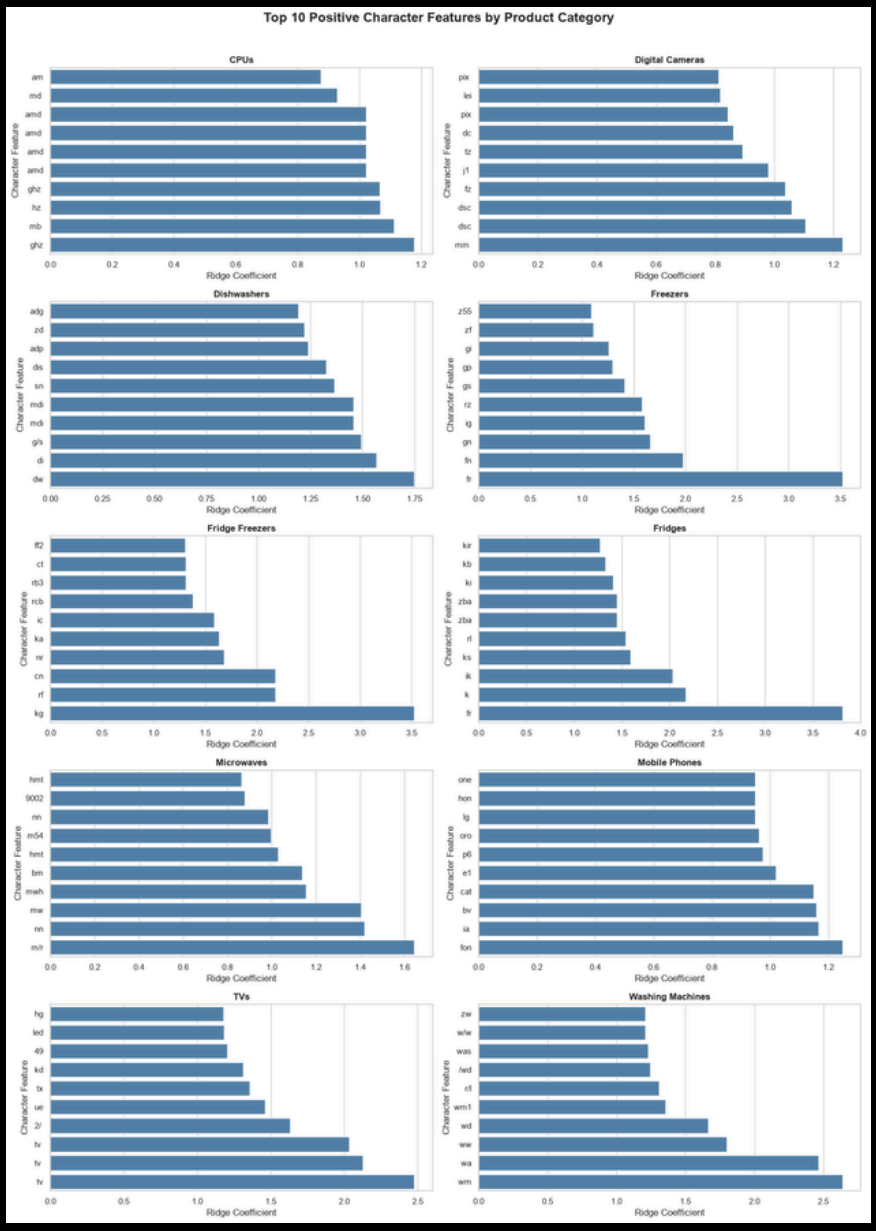
Explainability analysis was conducted to understand how the selected Ridge Classifier uses character-level patterns to distinguish product categories.

The analysis included:

- Category-specific positive and negative coefficients.
- Global character feature importance.
- Local feature contributions.
- Comparison of competing category scores.
- Mathematical validation of local explanations.



7.2 CATEGORY-SPECIFIC FEATURE IMPORTANCE



Explainability analysis was conducted to understand how the selected Ridge Classifier uses character-level patterns to distinguish product categories.

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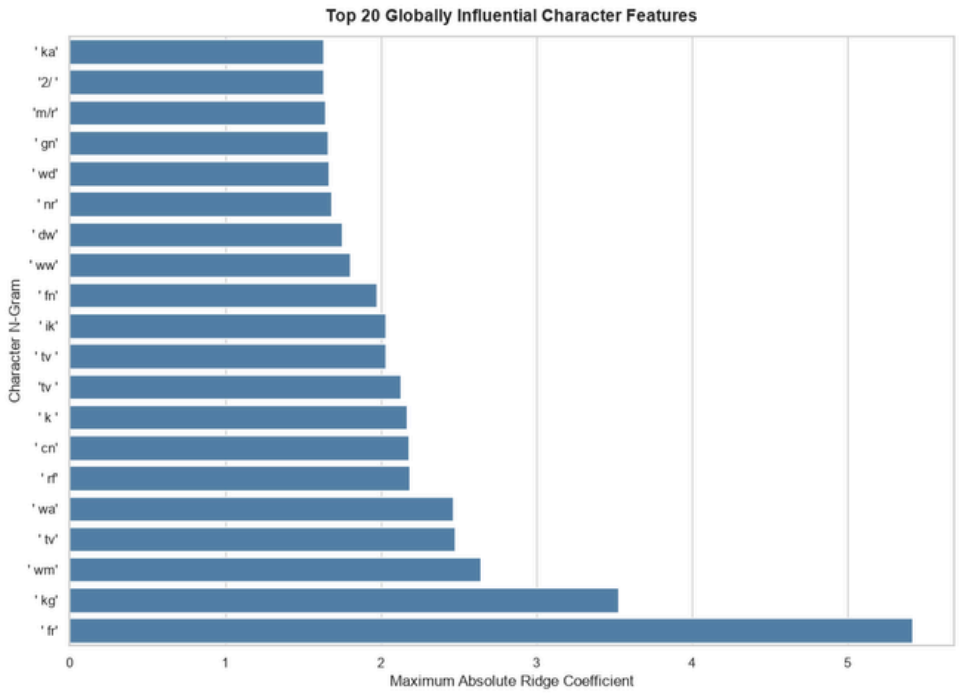
Examples included:

Category	Representative patterns
CPUs	intel, ghz, i7
Digital Cameras	**shot, canon fragments**
Dishwashers	**dish, sms**
Fridge Freezers	**kgn, idge**
Microwaves	**wave, hmt**
Mobile Phones	**hone, gb**
TVs	**tv, kd**
Washing Machines	**wash, kg**

Explainability analysis was conducted to understand how the selected Ridge Classifier uses character-level patterns to distinguish product categories.



7.3 GLOBAL FEATURE IMPORTANCE



Global importance was calculated using the maximum absolute coefficient across product categories.

Prominent character patterns included:

- fr - associated with refrigeration-related terminology.
- kg - associated with appliance specifications.
- wm - associated with Washing Machines.
- tv - associated with TVs.

The findings show that short character sequences can carry substantial predictive information.

However, coefficient magnitude reflects model influence rather than a causal relationship with product category.

7.4 LOCAL PREDICTION EXPLANATIONS

Correct Prediction - Mobile Phone

***Product**: iphone 6s space grey 128gb 4.7 unlocked sim free*

- Actual category: Mobile Phones.
- Predicted category: Mobile Phones.
- Mobile Phone decision score: 1.0065.
- CPU decision score: -1.0192.

The model identified mobile-phone-related character patterns and produced a clear separation between the two category scores.

Incorrect Prediction - Android Device

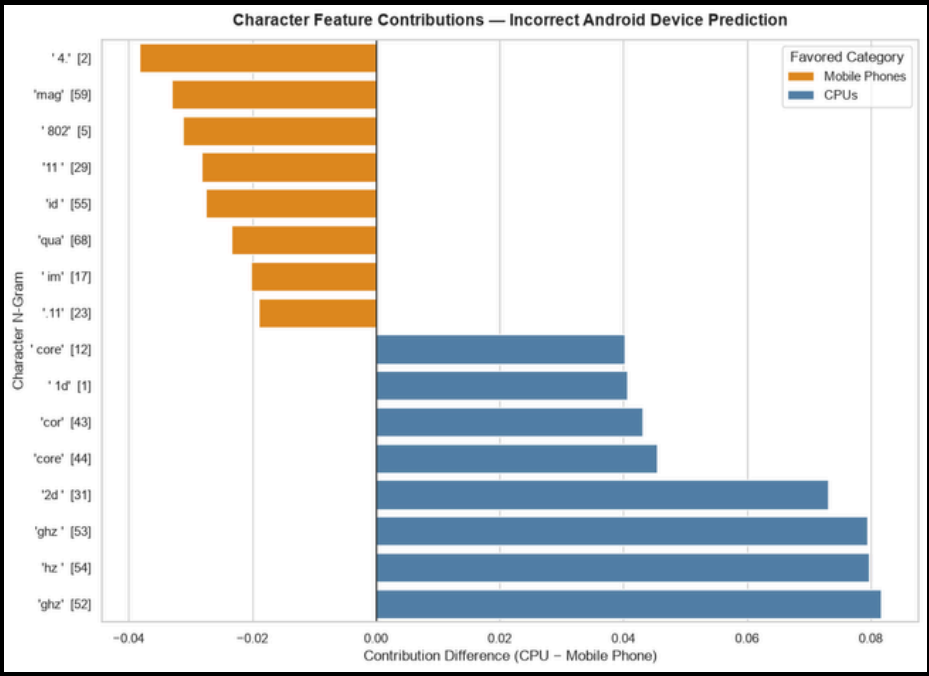
***Product**: android 4.4.4 802.11 a/b/g/n 1d/2d imager hi2d 1.2 ghz quad core*

- Actual category: Mobile Phones.
- Predicted category: CPUs.
- CPU decision score: -0.3830.
- Mobile Phone decision score: -0.3858.
- Decision margin: 0.0028.

The nearly identical category scores indicate a highly ambiguous prediction.



7.5 COMPETING FEATURE CONTRIBUTIONS



The analysis compared how individual character patterns contributed to the CPU and Mobile Phone scores.

Features favoring CPUs

- ghz: 0.0817
- hz : 0.0798
- ghz : 0.0795
- core: 0.0454

Features favoring Mobile Phones

- 4.: -0.0383
- mag: -0.0330
- 802: -0.0313
- 11 : -0.0282

The values represent differences between CPU and Mobile Phone feature contributions.

The competing evidence explains why the model narrowly favored CPUs despite the product belonging to Mobile Phones.

7.6 EXPLAINABILITY VALIDATION

Local explanations were validated by reconstructing model decision scores using:

Feature Contributions + Model Intercept = Decision Score

Prediction	Reconstructed score	Model score
Correct Mobile Phone	1.0065	1.0065
Incorrect CPU	-0.3830	-0.3830

Both reconstructed scores matched the original model outputs at the displayed precision.

This confirms the mathematical consistency of the local feature contribution calculations.

7.7 KEY EXPLAINABILITY FINDINGS

- Character-level features capture recognizable product terminology and technical specifications.
- Product identifiers and brand-specific naming patterns also influence predictions.
- Overlapping technical terminology contributes to ambiguous classifications.
- Local feature contributions provide a transparent explanation of individual Ridge Classifier decisions.

